

Qatar-9b

R-2451

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-9_250220 · created 2026-08-23T00:05:37+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460726.906 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.126 +/- 0.06
Transit depth (Rp/Rs)^2	1.6 +/- 1.5 [%]
Orbital Inclination (inc)	87.85 +/- 1.7
Airmass coefficient 1 (a1)	1.3 +/- 0.51
Airmass coefficient 2 (a2)	-0.28 +/- 1.35
Scatter in the residuals of the lightcurve fit is	37.7 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	3.75
Transit Duration (day)	0.037 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.126 $\pm$ 0.06 vs 0.1489 (deviation 0.25 σ)
Duration fit vs published	0.037 d vs 0.0779 d (deviation 2.46 σ)
Mid-time O-C	16.9 min
Depth SNR	1.4
Residual scatter	37.7 %

Light curve

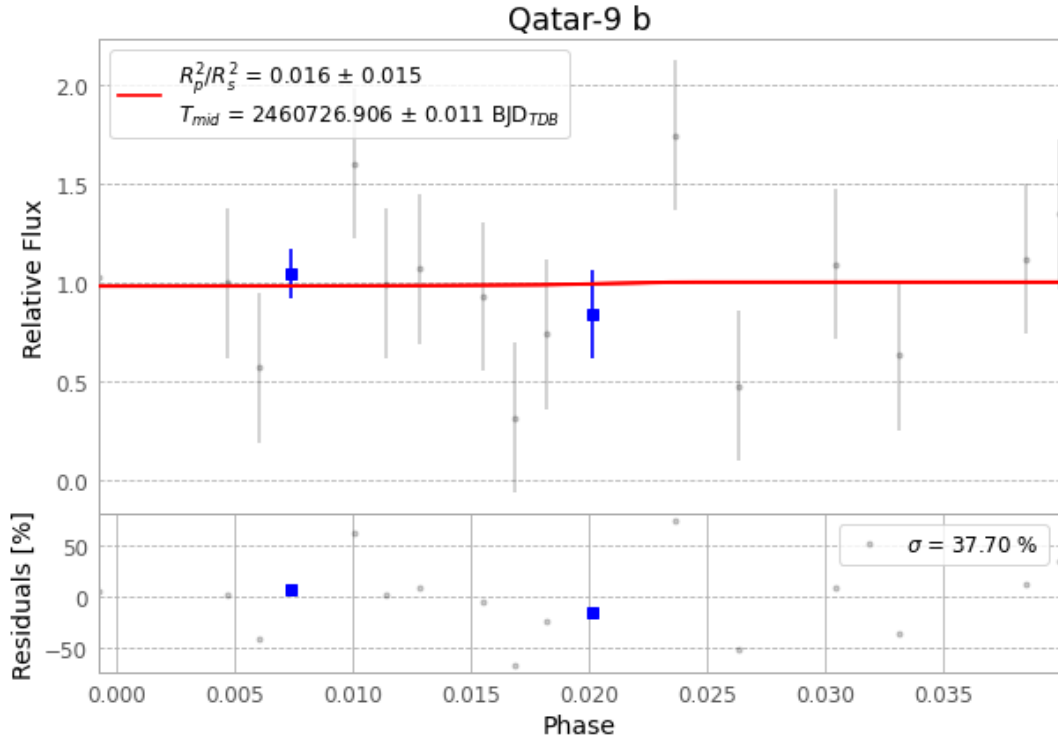


Figure 1 — FinalLightCurve_Qatar-9 b_2025-02-20.png (SHA-256 d6341448c963482e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [550, 340], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 25caf9b301efc1f3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

35 FITS frames (23 MB), Qatar-9250220093617.FITS → Qatar-9250220111815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-250220.log (7a5a1d6f49f6...), FinalLightCurve_Qatar-9 b_2025-02-20.png (d6341448c963...), FinalLightCurve_Qatar-9 b_2025-02-20.csv (bd5f633245b5...)

Compute resources

Wall time 77.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 00:05Z.